USN | | | | | | | | | | [22CSE216]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

CLOUD COMPUTING AND VIRTUALIZATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A			**5x2=10**
1.	**a.**	Differentiate Cloud Computing and Virtualization.	L2	CO1	**02**
	b.	To ensure the security of data in the cloud, how can it be secured while in transit and at rest?	L3	CO5	**02**
	c.	What is the role of data encryption in cloud storage?	L2	CO1	**02**
	d.	Outline the primary function of Azure Virtual Machines (VMs)?	L2	CO1	**02**
	e.	In terms of cloud computing, define virtualization.	L2	CO1	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Analyze the ways in which cloud computing is supported by virtualisation technology.	L4	CO3	**05**
		OR			
	b.	Demonstrate the need for a private cloud.	L3	CO5	**05**
3.	**a.**	What is the Cloud Resource Monitoring Process? Briefly explain the key areas of Monitoring.	L2	CO1	**05**
		OR			
	b.	Summarize the hybrid cloud with a use case in a few brief paragraphs.	L2	CO5	**05**
4.	**a.**	Differentiate Object Storage, Block Storage and File Storage.	L2	CO1	**05**
		OR			
	b.	Explain the concept of scalability in cloud storage, and discuss its two main forms along with their benefits.	L2	CO1	**05**
5.	**a.**	Evaluate the disaster recovery options available in Microsoft Azure.	L5	CO4	**05**
		OR			
	b.	Design a data storage strategy using GCP for a company handling large amounts of structured and unstructured data.	L5	CO4	**05**
6.	**a.**	Describe how cloud virtualization supports scalability and address two challenges that organizations run with when implementing virtualisation into practice.	L2	CO1	**05**
		OR			
	b.	Compare the key features of Docker and Kubernetes, and discuss how they work best together for cloud-native applications.	L2	CO1	**05**
7.	**a.**	Analyze the benefits and challenges of deploying Virtual Desktop Infrastructure (VDI) in the cloud,	L5	CO4	**05**
		OR			
	b.	Outline the main functions of cloud management tools and analyze how they make the management of virtualized environments easier.	L4	CO3	**05**

Learning Resource Centre Jain Global Campus Ramanagar-112

		PART - C			6x10=60
8.	a.	Describe each cloud computing service in detail. In each service category, list two cloud providers by company name.	L2	CO1	10
		OR			
	b.	"Cloud computing can save money,". What do you think? Which databases are available as open-source cloud computing platforms? Describe a single database in detail.	L2	CO1	10
9.	a.	Discuss the critical components that are necessary for the establishment of a successful cloud computing infrastructure.	L2	CO1	10
		OR			
	b.	Describe in detail Key Cloud Compliance Regulations and Standards.	L3	CO5	10
10.	a.	What is a Virtual Private Cloud (VPC)? Explain its architecture and key components. Discuss the benefits of using a VPC in cloud environments, and how it enhances security, scalability, and flexibility for businesses.	L2	CO1	10
		OR			
	b.	Explore the advantages and Challenges businesses face when using cloud storage. How can cloud storage enhance operational efficiency, and what security risks need to be addressed?	L3	CO5	10
11.	a.	Discuss the core services provided by Google Cloud Platform (GCP) and how they can benefit businesses in terms of scalability, cost-efficiency, and innovation with an example.	L2	CO1	10
		OR			
	b.	Analyze the cost optimization strategies offered by AWS, Azure, and GCP for controlling cloud expenses. Discuss how effective these strategies are in minimizing cloud costs for both short-term and long-term projects.	L4	CO3	10
12.	a.	Compare and contrast the global infrastructure and availability zones of AWS, Azure, and Google Cloud Platform (GCP), and analyze how these impact performance, reliability, and data redundancy.	L4	CO3	10
		OR			
	b.	Describe in detail on Amazon Web Service (AWS) providers.	L2	CO1	10
13.	a.	Analyze the various virtualization models (desktop, network, server, and storage) utilized in cloud computing.	L4	CO3	10
		OR			
	b.	Describe the architecture of virtual machines (VMs) in a cloud environment, and critically analyze their advantages and limitations.	L4	CO3	10